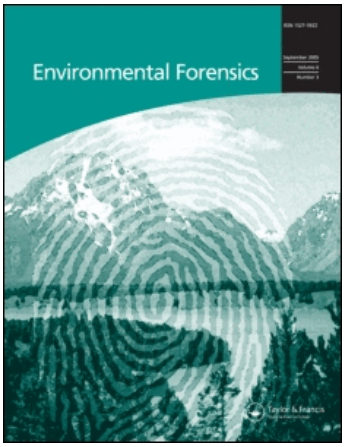
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Forensic Fingerprinting of Oil-Spill Hydrocarbons in a Methanogenic Environment–Mandan, ND and Bemidji, MN

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In recent decades forensic fingerprinting of oil-spill hydrocarbons has emerged as an important tool for correlating oils and for evaluating their source and character. Two long-term hydrocarbon spills, an off-road diesel spill (Mandan, ND) and a crude oil spill (Bemidji, MN) experiencing methanogenic biodegradation were previously shown to be undergoing an unexpected progression of homologous *n*-alkane and *n*-alkylated cyclohexane loss. Both exhibited degradative losses proceeding from the high-molecular-weight end of the distributions, along with transitory concentration increases of lower-molecular-weight homologs. Particularly in the case of the diesel fuel spill, these methanogenic degradative patterns can result in series distributions that mimic lower cut refinery fuels or admixture with lower cut fuels. Forensic fingerprinting in this long-term spill must therefore rely on more recalcitrant series, such as polycyclic aromatic hydrocarbon or drimane sesquiterpane profiles, to prove if the spilled oil is single-sourced or whether there is verifiable admixture with other extraneous refinery fuels. Degradation processes impacting *n*-alkanes and *n*-alkylated ring compounds, which make these compounds unsuitable for fingerprinting, nevertheless are of interest in understanding methanogenic biodegradation.

Keywords: fingerprinting, Mandan, ND, Bemidji, MN, oil spill, methanogenic biodegradation

Introduction

Forensic fingerprinting is widely used to characterize, correlate or differentiate, monitor, and establish sources of oils, oil spills, and other hydrocarbon entities (Wang and Fingas, 2003; Stout et al., 2002). Fingerprinting involves a rigorous analysis of chemical patterns and diagnostic relationships between specific chemicals within each oil. All oils have their own distinctive chemical fingerprint. However, once released into the environment petroleum hydrocarbons are immediately subject to effects of physical and chemical weathering and microbial biodegradation. These changes, both degree of alteration and progression of the loss, can significantly impact the fingerprint and thus need to be taken into consideration.

Pathways of aerobic weathering and degradation have been widely studied and the degradation progression well established. The products of aerobic degradation are frequently polar compounds with oxygen functional groups. Examples are organic acids, esters, alcohols, and related compounds (Wilkes et al., 2002). General loss of chemical constituents from spilled petroleum hydrocarbon proceeds with loss first of *n*-alkanes (light degradation), then isoprenoids, cyclic alkanes, and lower-molecular-weight (MW) aromatics (moderate degradation), followed by the remainder of the more complex, higher MW constituents (heavy to severe degradation) (Peters and Moldowan, 1993). This order has been further refined (Kaplan et al., 1997; Wang et al., 1998) and applied in forensic case studies. The order for *n*-alkane loss is reported to be: from low-range, to mid-range, and then finally to high-range carbon number homologs. Other related homologous series, i.e., ring compounds with *n*-alkyl side chains, include *n*-alkylcyclohexanes (CHs) and *n*-alkyl benzenes. The *n*-alkyl CHs are more refractory than *n*-alkanes and persist in the oil environment for a longer time. Because their distributions carry the signature of the spilled oil or oil product longer, they have been utilized to differentiate fuel types, with characteristic homolog distributions shown to occur with different refinery products like kerosene, diesel fuel, and others (Kaplan and Galperin, 1996; Kaplan et al., 1997). The geochemical literature does not track the sequential biodegradation loss within the suite of *n*-alkyl CHs. However, in the case of *n*-alkylbenzenes, Kaplan et al. (1997) stated that the longer

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the alkyl chain attached to the aromatic ring, the more resistant the hydrocarbon is to biodegradation.

Unlike aerobic biodegradation, anaerobic biodegradation, in general, has been slower to be recognized. However, anaerobic biodegradation of hydrocarbons has been the subject of lively discussion in recent years (e.g., Fukui et al., 1999; Spormann and Widdel, 2000; Rueter et al., 1994; Zengler et al., 1999; Anderson and Lovley, 2000; Wilkes et al., 2003; So et al., 2003) and knowledge of its progression, mechanisms, intermediates and products is emerging. Anaerobic processes involving aromatic hydrocarbons such as benzene, toluene, ethylbenzene, and xylene (BTEX compounds) and other alkylbenzenes are better known than those of aliphatic hydrocarbons (Spormann and Widdel, 2000; Wilkes et al., 2000; Martus and Püttmann, 2003). In spite of the fact that *n*-alkanes frequently are the dominant constituents in spilled oils and fuels, they are considered to be relatively chemically inert. Although their degradation by sulfate-reducing bacteria was reported as early as 1944 (Novelli and ZoBell, 1944), only relatively recently has their susceptibility to anaerobic degradation been intensively investigated (Aeckersberg et al., 1991; Rueter et al., 1994; Zengler et al., 1999; Anderson and Lovley, 2000; Caldwell et al., 1998). Identified products of anaerobic biodegradation have thus far been primarily polar oxygenated compounds such as organic esters (fumarates, succinates) and acids (e.g., Wilkes et al., 2002)

Documentation of the specific progression of biodegradative losses within homologous families, e.g., long chain *n*-alkanes and *n*-alkyl substituted ring compounds, under anaerobic conditions has been rare. The commonly found preferential loss of lower MW *n*-alkanes from a crude oil has been shown in one laboratory study to occur under anaerobic sulfate-reducing conditions (Wilkes et al., 1995). Another study of artificially weathered oil under sulfate-reducing conditions shows a very small preferential loss of low (n-C_{17}) over high (n-C_{30}) MW alkanes and essentially equivalent losses of n-C_{17} and n-C_{30} under methanogenic conditions (Townsend et al., 2003). In a study of crude oils in reservoirs impacted by anaerobic biodegradation, George et al. (2002) found no evidence that lower MW aliphatic hydrocarbons (e.g., C_5) are depleted faster than higher MW analogs (C_9).

Both the Bemidji, Montana, and the Mandan, North Dakota site in our study are at stages of methanogenic biodegradation where the *n*-alkanes go from prominent to nearly or completely depleted, and *n*-alkyl CHs, after an initial lag due to their more refractory character, are also diminished (Hostettler and Kvenvolden 2002; Hostettler et al., 2001). However, in contrast to the aforementioned studies, both *n*-alkyl homologous series at Mandan and Bemidji show preferential losses from the high-MW end of the distribution and transitory enhancement at the low end. Aspects of this phenomenon have been noted only a few times in the literature. George et al. (2002) showed that terminal methyl groups on aliphatic hydrocarbons in crude oils are more prone to bacterial attack compared to mid-chain isomers. This supports the concept that degradation favors the ends of straight chains. Kim et al. (2005) suggest that degrading enzymes attack long alkyl side chains on NSO (nitrogen, sulfur, oxygen) compounds as if they were alkanes. In an article on diesel fuel degradation, Stout and Lundegard (1998) noted a progression of *n*-alkane loss from the high-MW end, and a small enhancement at the low end was also observed. The degradation was labeled as intrinsic biodegradation and the mechanism was suggested to be "shortening" of the alkyl chains via microbial (enzymatic) cleavage of the terminal carbon. However, no mechanism supporting the process suggested in this work or in our studies has yet been documented.

Because this anomalous progression, particularly with its enhanced concentrations of lower end homologs, might be construed as extraneous admixture with other hydrocarbon entities, understanding how degradative processes impact the spilled hydrocarbons in these two methanogenic systems is essential in evaluating the oil's forensic fingerprint. Only those components that are refractory, that is, not actively being changed, or are co-varying systematically, can be used for the fingerprint. When significant alteration to compounds or families of compounds in a spilled oil is occurring, Wang and Fingas (2003) noted that it is necessary to use relatively more degradation-resistant compounds such as polycyclic aromatic hydrocarbon (PAH) and biomarker compounds to fingerprint the oil. It is the responsibility of the forensic investigator to recognize these effects and account for them in their investigation (Stout et al., 2002). Therefore, although specific compounds that are being depleted can give information about processes impacting the oil, their usefulness for the forensic fingerprint is mitigated by their changing profiles. In the following study we contrast the specific oil fingerprint at Mandan using refractory components such as PAH and drimane biomarkers with the changing *n*-alkane and *n*-alkyl CH and benzene ring series. The fingerprint provides source, fuel product determination, and correlation information, and the changing profiles contribute new information on the methanogenic biodegradation process.

Field Sites and Background

The field studies include two long-term oil spills. These spills have been described previously (Bekins et al., 2005, and references therein; Hostettler et al., 2001; Hostettler and Kvenvolden, 2002, North Dakota) and are summarized here. Both spills have been constrained within subsurface shallow sandy aquifers for over 20 years. The first site involves an oil spill involving a single release of crude oil from a ruptured pipeline at Bemidji, Montana, in 1979. This spill introduced crude oil into a pristine environment where there were no confounding hydrocarbons to interfere with the oil signature. The spilled oil was a light paraffinic crude oil from Alberta. The Bemidji site has been extensively studied. Methanogenic conditions have been well documented within the main oil body, including data from 1997 (Chaplin et al., 2002) to the present (Bekins et al., 2005). Representative oil samples taken from different core locations within the oil body (Figure 1A, Table 1) are at varying stages of degradation due to conditions discussed in Bekins et al. (2005). Analysis of samples at these varying stages, some taken in different years,

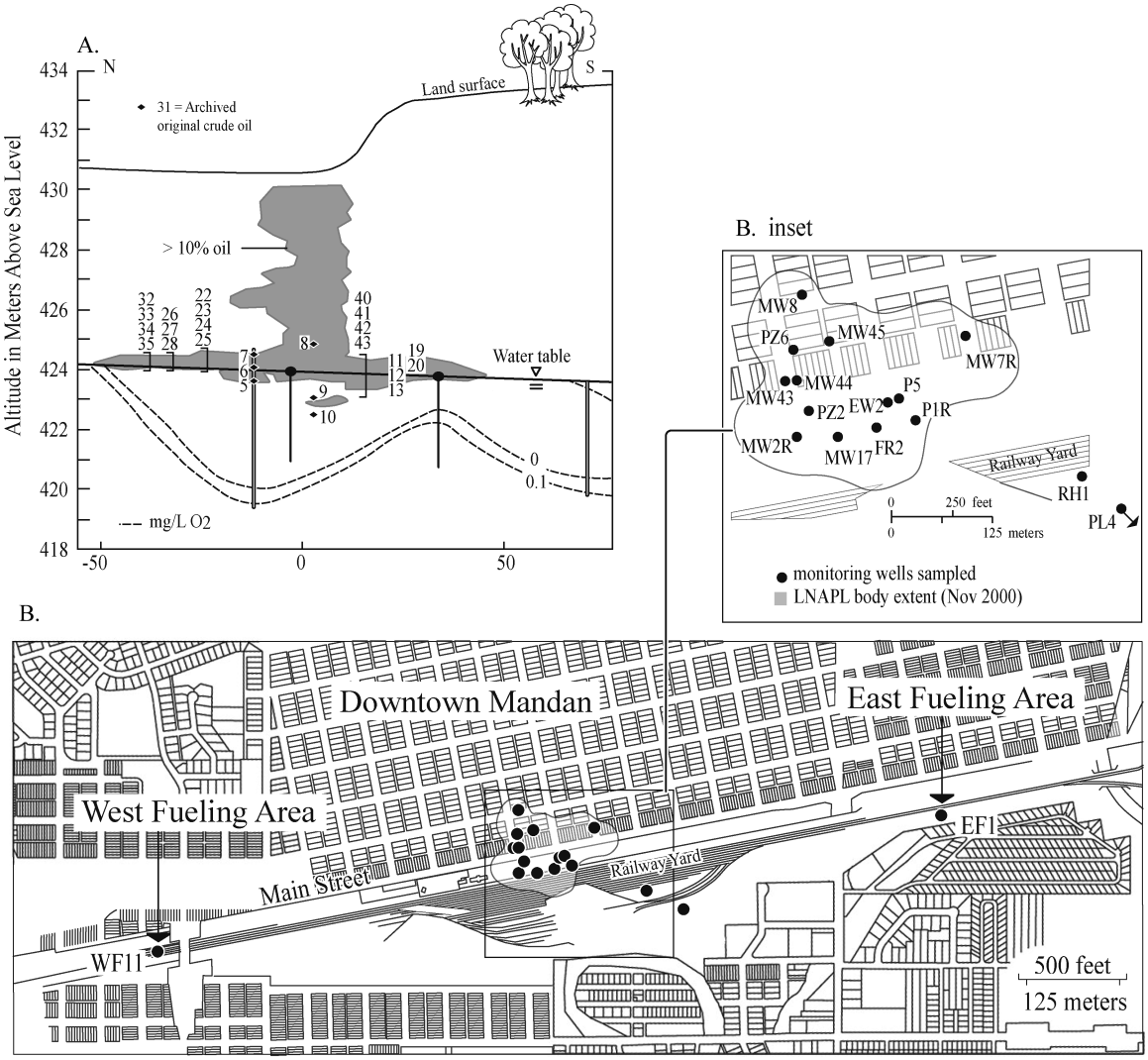


Figure 1. Diagrams of oil spill accumulations. A) Bemidji, Montana, spill (adapted from Bekins et al., 2005). B) Mandan, North Dakota, downtown spill location (adapted from Hostettler et al., 2001).

along with some archived core samples, allows one to track degradation progression. The pristine nature of the spill environment allows discrimination of strictly biodegradative changes since there is no possibility of oil co-contamination. The oil has been shown to be stationary within the sandy soil matrix and not subject to further migration. In both of these field studies the analysis focuses on the non-polar constituents retained within the oil body.

The second spill is a large hydrocarbon accumulation of light non-aqueous phase liquid (LNAPL) floating on the water table about 20 feet below land surface in Mandan, North Dakota. The LNAPL was postulated to be primarily diesel fuel (off-road diesel) from chronic spillage (1953–1991) at a local railway-refueling depot (Hostettler et al., 2001; Hostettler and Kvenvolden, 2002). The downtown LNAPL accumulation is contiguous with spillage from the railroad refueling area where the railroad received, stored, and transported the millions of gallons of diesel fuel per year that it used for more than three decades at this location. The estimated volume of this spill is 1.5 to 3.0 million gallons. Subsequent additional spillage occurred at two outlying refueling depots, a broken pipeline, and near an old railroad roundhouse site (Roberts, 2000). Figure 1B shows the extent of the downtown spill and indicates the other areas of spillage. The timeline of the spillage and the samples collected are reported in Table 2. The immediate environment of the spill is highly methanogenic (Breyer and Cowart, 2003). The more recent spills and the chronic nature of the downtown spillage provided samples at different stages of degradation, which allowed a degradation progression to be observed, as with Bemidji.

The diesel fuel used by the railroad at this site has historically been supplied by a single local refinery, which has continuously utilized crude oil from the nearby Williston Basin. A sample of the diesel, RR40, supplied in November 2000, is a refinery off-road, non-desulfurized diesel fuel typical of fuel in use over this entire time period and is used as a reference standard. This

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Table 1. *Bemidji, MN, samples from this study (see figure 1A)*

	State[a]	
BE- #	Degradation	Core[b]
6, 7	D,D	0116
8	D	0119
15, 16, 18	A,B,C	9604[c]
19	A	9614[c]
22, 23	D,D	9709[c]
27, 28, 29	D,C,B	9710[c]
31	A	Original oil[c]
32, 35	D,C	0219
36	A	0227
40, 41	B,B	0232

[a]Defined in text: Range from: A (least degraded) to D (*n*-alkanes gone).
[b]First two numbers indicate year of core collection.
[c]Frozen, archived.

fuel was blended to railroad specifications and has remained the same (cetane rating 40) over the time period of railroad usage (Roberts, 2001, personal communication).

At the request of the North Dakota Department of Health (Bismarck, ND) in 2000 our laboratory analyzed samples of the LNAPL to determine if the downtown spill is off-road diesel and/or whether contaminants other than railroad diesel are present. The samples were a very complex mixture of older degrading hydrocarbons from over 50 years of underground accumulation with succeedingly more recent hydrocarbons. Our report (Hostettler et al., 2001) concluded that spillage throughout the railroad yard and beneath the downtown was off-road diesel. No extraneous industrial or commercial chemicals or other hydrocarbon fuel products were found, with one exception. A trace of iso-octane, a marker for gasoline, was found in one shallow outlying downtown well, MW8, possibly from a very small local gasoline spill. The chemical analyses of the entire sample set showed minor differences in the relative concentrations of selected constituents within the LNAPL samples that the report attributed to probable small chemical differences in the refinery feedstock over the 50 years of local railway usage. That the diesel fuel has consistently been the same product over this whole time period, even though variably degraded, was born out by the narrow range of bulk carbon isotope values of the LNAPL samples and the RR40, a common red diesel dye marker, as well as the consistent profiles of refractory PAH constituents. The methanogenic biodegradation noted above that impacts *n*-alkanes and *n*-alkyl CHs, although anomalous and very interesting, was considered to render these homologous series unsuitable for forensic fingerprinting.

The railroad initially acknowledged the spillage and built a facility onsite that had recovered approximately 0.5 million gallons of LNAPL by 2000 (Roberts, 2000; Burlington Northern Santa Fe Railway, 2000a, 2000b). However, when litigation to enforce remediation became imminent in 2003, the claim that the downtown LNAPL was railroad diesel was disputed by the railroad. The railroad scientists and lawyers claimed at various times that the downtown LNAPL was not from the railroad (news report) or, alternatively, if it was railroad diesel, it was substantially mixed with other non-railroad hydrocarbon fuels (unpublished reports to BNSF, Stout et al., 2006; Stout and Uhler, 2006). This latter claim was based mainly on the *n*-alkyl CH profiles. In contrast to the low-MW homolog enhancement suggested in Hostettler and Kvenvolden (2002) to be a direct result of the biodegradation progression, the challenge from the railroad scientists claimed that the enhanced levels of *n*-alkyl CHs were due to admixture with other lower cut refinery products, specifically kerosene. The purpose of this present article is to give a more complete exposition of the forensic fingerprint of the Mandan LNAPL, with substantiating data from the Bemidji oil spill, and to look at the biodegradation progression of the *n*-alkylated compounds.

Laboratory Incubation Experiment

In addition, an independent laboratory experiment following methanogenic biodegradation was undertaken. A crude oil from the North Sea was incubated using activated sewage sludge as a methanogenic culture. Subsamples were collected over a two-year period and the progression of the *n*-alkane losses was determined.

Table 2. *Time-line of sample spillage, Mandan, ND*

Estimated dates of spillage	Samples	Description
1953–1984	EW2, FR2, FR4, MW2R, MW7R, MW8, MW17, MW43, MW44, MW45, MW45B, MW45P, P1R, P5, PZ2, PZ6	Downtown accumulation. Diesel refueling area at downtown RR yard; spillage noted from early 1950s. Extent of plume mapped in Roberts, 2000; Hostettler et al., 2001. Fueling moved to remote areas, 1984
1950s or 1960s	RH1	Roundhouse spill (old spill, mid-RR yard)
~1978–1984 to 1991	EF, WF11, WF11A	East/West fueling stations built and on-line; end of downtown pumping. Catchment trays installed to prevent further spillage at E/WF in 1991
Unknown start date to 1989/1990	PL4	Pipeline spill, delivered diesel directly from refinery
2000 (not spilled)	RR40, Reference oil	RR (off-road) diesel, blended to the same RR specifications from the same refinery, which took its feeder stock from Williston Basin crude oil, over the entire time of the RR usage

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Methods

Oil Spill Procedures

Procedures used to collect and analyze samples of petroleum hydrocarbons from the Mandan and Bemidji sites are detailed elsewhere (Hostettler and Kvenvolden 2002; Wang and Huang 2003). Briefly, Mandan samples were recovered as whole oils from belt driven pumps in wells extending to the floating LNAPL. Bemidji crude oil samples (except for the archived oil) were extracted from oil-contaminated sediments (>10% oil saturation) with dichloromethane (DCM). The Bemidji spill environment was pristine; the sediment is mostly sand with very low organic carbon and no confounding organic constituents. A simple DCM extraction was found to reproducibly extract the whole oils with no constituent discrimination. Similarly, no changes occurred when a sample of the original oil was mixed in triplicate with uncontaminated sediment and extracted with DCM. An archived sample of pre-spill Bemidji oil (BE-31) and a contemporary sample of unaltered railroad diesel (RR40) were used as reference standards for comparison with the biodegraded samples to evaluate the biodegradation progression.

The extracted hydrocarbons were analyzed by gas chromatography/mass spectrometry (GC/MS) either as whole oils dissolved in DCM or after being separated into aliphatic and aromatic fractions by liquid chromatography on a mixed silica-alumina column. GC/MS analysis of the Mandan samples utilized a lower range temperature program than was used for the crude oil samples because of the lower boiling range of diesel fuel relative to crude oil. Extracted ion chromatograms of m/z 57, 83, and 91 were used to profile *n*-alkanes, *n*-alkyl CHs, and *n*-alkylbenzenes respectively. Compounds in the respective homologous series were identified by comparison with known standards. A second long-chain alkyl benzene series was monitored at m/z 133 and tentatively identified as homologous fragments of isorenieratane by comparison with literature mass spectra from Koopmans et al. (1996). Another C_4 benzene (C_4Bz) was monitored at m/z 119, along with a sesquiterpane series including drimane at m/z 123. These compounds were used to evaluate source and possible admixture of fuels.

Compound-specific carbon isotope analysis (CSCIA), performed at Brown University (Providence, RI), was used to characterize the *n*-alkane constituents. For the CSCIA all the Bemidji and Mandan samples were grouped into four progressive stages of degradation: (A to D): A) little or no degradation, *n*-alkanes dominant (includes original unaltered oil from Bemidji and RR-40, the unaltered diesel standard equivalent to that used over time by the railroad for Mandan); B) light degradation, *n*-alkanes diminished; C) moderate degradation, only low-end *n*-alkanes prominent; and, D) most advanced degradation, some low end *n*-alkanes present in the Mandan but not the Bemidji samples; isoprenoids are dominant. The *n*-alkane distributions of three representative samples from each stage were then profiled by CSCIA in order to obtain a clearer picture of the biodegradative changes occurring in this compound class. First, the aliphatic fractions of these Mandan and Bemidji oils were subjected to urea adduction in order to isolate the *n*-alkanes. The urea adduction procedure has been described previously (Pond et al., 2002). Each hexane fraction containing the *n*-alkane isolates was checked by gas chromatography and then profiled by CSCIA using a gas chromatography-combustion-isotope ratio mass spectrometer (GC-C-IRMS). An HP 6890 GC (Agilent Technology Inc., Santa Clara, CA) was connected to a Thermo Finnigan Delta$^+$-XL mass spectrometer (Thermo Fisher Scientific, Waltham, MA) via the combustion interface. Six pulses of CO_2 reference gas with known $\delta^{13}C$ values were injected via the interface to the IRMS for the determination of $\delta^{13}C$ values of the sample compounds. Duplicate runs of each sample were made and the values for each *n*-alkane peak averaged. Standard deviations were small and ranged from 0 to 0.30/00.

Semi-quantitative analysis by GC/MS of the *n*-alkanes within both groups of oils was based on the ratio of individual hydrocarbons to a conserved internal standard. A bicyclic sesquiterpane, 8β(H)-drimane, was used for normalizing the *n*-alkane and *n*-alkyl cyclohexane profiles in the Mandan oil samples because this compound was found to remain constant relative to refractory PAH in the samples (Hostettler and Kvenvolden, 2002). The quantitation internal standard d_{10}-acenaphthene (m/z 164) was used for normalizing *n*-alkylbenzenes. Pristane (Pr) was used in the case of Bemidji oil samples after it was found to be unaltered with respect to hopane biomarkers.

Bulk stable carbon isotope analysis for the whole oil samples was performed by Zymax Corp. (San Luis Obispo, CA). High-temperature gas chromatography analysis on unaltered Bemidji oil was done by Integrated Laboratory Technologies (Richmond, CA). Ambient air samples from wells in the downtown Mandan spill area were collected in April 2005 by personnel of Legette, Brashears & Graham, Inc. (St. Paul, MN), for methane headspace and stable carbon isotope analysis. Headspace analysis was done by Jennifer Dougherty of the United States Geological Survey (Menlo Park, CA), and CSCIA of the methane by Paul Eby, University of Victoria, Canada.

Laboratory Incubation Experiment

Anaerobic biodegradation of a North Sea crude oil (Bonnie Light, Chevron) was conducted in butyl rubber-sealed serum bottles. Each bottle contained 0.6 mL of activated sewage sludge (3% solids by mass), which was collected in a sterile glass bottle from the municipal wastewater treatment plant in South Bend, Indiana. Sludge (0.6 mL) was added to each of fifteen 60-mL serum bottles containing anoxic mineral medium (20 mL, pH 7, 0.033 M K_2HPO_4, 0.033 M KH_2PO_4, 0.015 M $(NH_4)_2SO_4$, 1.2 mM $MgCl_2$, 0.18 mM $CaCl_2$, 0.0097 mM $NaMoO_4$, 0.0066 mM $FeSO_4$, 0.012 mM $MnCl_2$). Bonnie Light crude oil was pipetted into each serum bottle (final concentration 1% v/v) and distributed by brief shaking, which caused hydrocarbon droplets to associate with sludge particles. The bottles contained an anoxic headspace (one third or less of bottle volume) initially gassed with N_2. Killed controls were created by autoclaving the bottles (121°C, 15 psi for 20 min). After being autoclaved, the bottles were set aside for 48 h to permit germination of viable endospores and were autoclaved again. Bottles (live cultures

and killed controls) were incubated in the dark at 37°C. Three individual bottles, duplicate live cultures and a killed control, were collected on days 0, 150, 300, 570, and 730. The collected samples were held at 4°C while shipped overnight to Brown University for GC analysis. Qualitative headspace analysis by GC/MS verified methane production in the live samples during the course of the experiment.

Sample Preparation

Culture subsamples from the laboratory tests were subjected to extraction, separation, and GC analysis. Each oil and sludge subsample was transferred into 40-mL Pyrex glass tubes and promptly extracted from the microbes with hexane to prevent further degradation. Each extraction involved adding 2 mL of hexane to the sample, centrifuging the sample for 10 min at 1500 rpm, and pipetting the hexane/hydrocarbon layer from the sample tube. Each sample was extracted three times, and all of the hexane fractions from each sample were collected in a single vial.

Oil hydrocarbons for all samples were then separated into fractions by flash-column silica gel (60 Å 230–400 mesh) chromatography. Two fractions were collected for subsequent GC analysis; the first was eluted with hexane, and the second was eluted with DCM.

Gas Chromatography

Quantification of *n*-alkanes was conducted using an Agilent HP 6890$^+$ gas chromatograph, fitted with a 60 m fused silica column (HP-1MS, 0.32 mm i.d., 0.25 μm film thickness), a flame ionization detector, a split/splitless injector, and an HP 7683 series auto-sampler (Agilent Technology, Inc., Santa Clara, CA). Helium (UHP 5.0 grade) was used as the carrier gas, operating in constant flow mode at a rate of 1.7 mL.min^{-1}. The temperature program for *n*-alkanes was isothermal at 40°C for 1 min, followed by heating to 315°C at 5°C·min^{-1}, and held for 30 min. The injector was operated at a split ratio of 5:1. Hexane was used as solvent to dilute samples for GC analyses. Concentrations of the compounds were determined by using the area-quantity relationship established from six external standards (C_{21}, C_{23}, C_{25}, C_{27}, C_{29}, and C_{30} *n*-alkanes). Quantification of *n*-alkanes was based on the ratio of individual compounds to a conserved internal standard. Pristane was used after it was found to be unaltered with respect to hopane biomarkers, as was done in the previous study of aerobic biodegradation (Pond et al., 2002). Culture samples from two duplicate experiments and a killed control at each time period were quantified for *n*-alkanes.

Results and Discussion

Biodegrading Constituents

n-*Alkanes*

Although the forensic interpretation and challenge focused on *n*-alkyl CHs, it is necessary to examine the *n*-alkane series in order to see how the biodegradation progression impacts straight chain homologous series. Chromatograms illustrating the progression of methanogenic biodegradation of *n*-alkanes are given for both the Mandan and Bemidji samples in Hostettler et al. (2001) and Hostettler and Kvenvolden (2002). The *n*-alkane progression from the laboratory incubation study will be discussed later. For further visualization, *n*-alkane homologs from a set of six representative Mandan LNAPL (whole oil) samples covering the degradation range were normalized to the conserved internal standard in order to facilitate comparison between samples. The samples were then placed in order of apparent degradation, according to sequential loss of *n*-alkanes, and the *n*-alkanes graphed in Figure 2. Here it can be seen that higher MW *n*-alkanes are disappearing first. Bulk stable carbon isotope values are given for each sample on its respective plot. The range of bulk stable isotope values for the complete set of Mandan LNAPL samples is −29.52 to −28.30‰ (from unaltered to most-degraded sample), yielding a differential of 1.22‰ (Hostettler et al., 2001). Carbon isotope values for methane in well headspace collected in 2003 are also given and ranged from −56.0 to −41.7‰. The bulk stable carbon isotope values of the residual oil at Mandan became progressively isotopically heavier, that is, enriched in ^{13}C, and the methane is significantly lighter, as would be expected in a biodegrading system. The bulk carbon isotope values of the residual oils, then, track *n*-alkane biodegradation in the oil bodies. The observed progression of preferential higher-end *n*-alkane loss is quite different from that cited for other systems (e.g., Wang et al., 1998; Kaplan et al., 1997; Kaplan and Galperin, 1996; Wilkes et al., 1995) where low-end *n*-alkanes disappear first. No attribution of aerobic or anaerobic conditions is included in these earlier works. Thus, this present study suggests that a different mechanism of biodegradation or different microbial substrate utilization occurs in methanogenic environments.

In addition, it is apparent in Figure 2 that there is an absolute increase of lower-end *n*-alkanes over and above initial concentrations, that is, concentrations in the reference sample RR40. In the Mandan progression shown in this figure the degrading diesel fuel has enhanced concentrations particularly notable at the *n*-C_9 to *n*-C_{12} range. These concentrations can be up to more than 1.7 times the original concentration in RR-40. In the example from previous work (Hostettler and Kvenvolden, 2002), abundances of *n*-C_{11} are 57%, 73%, and 26% higher than RR-40 in LNAPL samples WF11, MW44, and MW17, respectively. The pattern of *n*-alkane loss also progresses with a shift of the C_{max} in the distribution to lower MW homologs.

Compound-specific carbon isotope analysis (CSCIA) sheds more light on these excess concentrations and is a useful tool to help evaluate the concentration enhancements. CSCIA provides a much more sensitive indicator of the degree of oil degradation than bulk δ^{13}C values given above. This technique focuses entirely on the primary degrading aliphatic oil constituents, the *n*-alkanes, especially pinpointing which homologs are being impacted by degradation. CSCIA chromatograms of *n*-alkanes from C_{11} to C_{23}, displayed for simplicity as average values for

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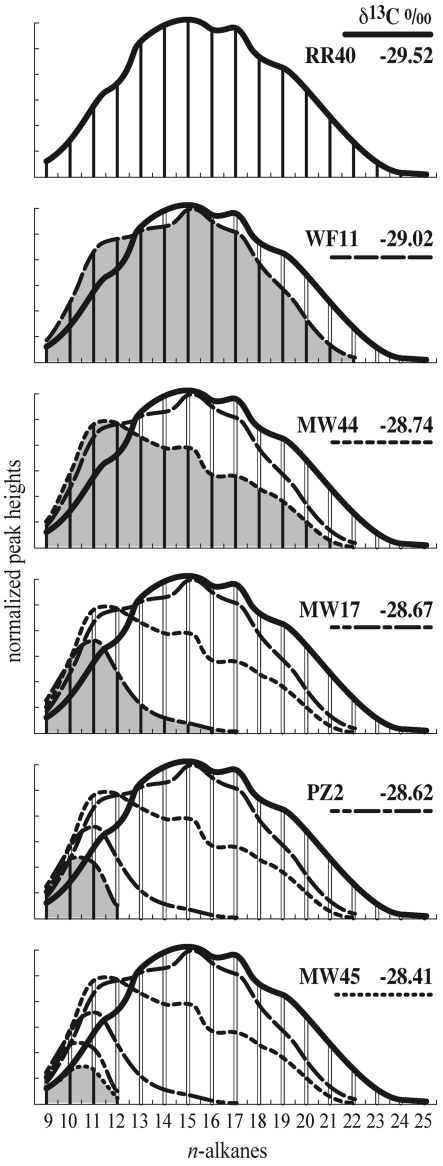


Figure 2. *n*-Alkane profiles of six representative Mandan, North Dakota, oil samples. Peaks from gas chromatography/mass spectrometry analysis are normalized to a conserved internal standard so that all samples are comparable. Samples are placed in order of progressive overall loss of *n*-alkanes. The reference standard RR40 is non-degraded and the four progressive stages are represented by WF11 (A), MW44 (B), MW17 and PZ2 (C), and MW45 (D). Bulk δ ^{13}C values are given for each oil. Well headspace δ^{13}C values ranged from −56.0 to −41.7‰.

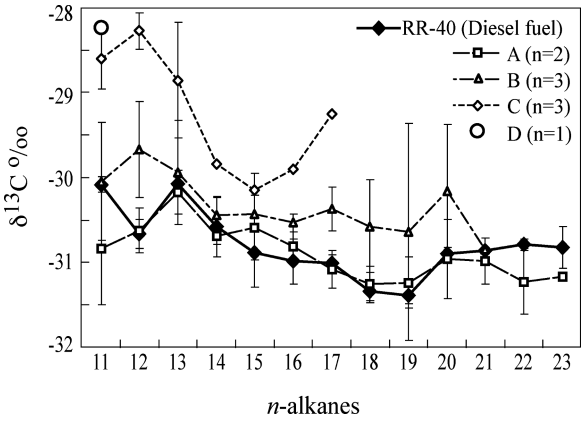


Figure 3. Compound-specific carbon isotope profiles of *n*-alkanes in representative Mandan, North Dakota, oil samples. Profiles for the four progressive ranges of biodegradation, A–D (see text) were averaged for simplicity. Only one of the analyzed D-samples had enough *n*-alkanes to measure.

each of the four degradation groups, are shown in Figure 3. The lightly degraded A-group samples show very little difference from the RR-40 standard. The B-group shows some enrichment in ^{13}C at mid-range, approximately *n*-C_{17} to *n*-C_{20}. However, the C-group, which has lost most of its upper range *n*-alkanes, shows a significant ^{13}C enrichment in its homologs, especially in the homologs (*n*-C_{11} and *n*-C_{12}) that were shown in Figure 2 to have enhanced concentrations. One explanation for this trend could be that higher end *n*-alkanes are sequentially losing terminal carbons to form lower end *n*-alkanes. As a portion of the constituents of a specific carbon number are lost to methanogenesis, the portion of the *n*-alkanes of that same carbon number that is not yet impacted would be expected to exhibit enrichment in ^{13}C versus ^{12}C (George et al., 2002). Moreover, the enhanced absolute concentrations, along with the enrichment of ^{13}C at those carbon numbers, implies that a portion of the enhanced concentration is a secondary product (i.e., shorter *n*-alkanes formed after loss of a lighter methane or methane-producing unit). Therefore, both enhanced concentrations and their heavier isotopic values support the explanation that a significant proportion of low-range *n*-alkanes at later stages of alkane biodegradation are secondary products derived from upper range *n*-alkanes.

The Bemidji sample set was also separated into four progressively biodegraded groups. Bulk carbon isotope analysis also tracks this degradation, with a range of −28.85 to −28.44‰ (a smaller differential than with Mandan because of the greater complexity of crude oil versus diesel fuel) and a range of methane values of −42.05 to −33.35‰ (Revesz et al., 1995). A possible homolog enhancement and isotopic enrichment is also seen in this sample set, although progression of *n*-alkane degradation is more subtle due to the broader range of *n*-alkanes in a crude oil versus diesel fuel. For example, BE-31 is unaltered and group A is lightly altered, but a difference in the *n*-alkane profile is beginning to be noticeable and is observable in a comparison of BE-31 to a group-A oil, BE-19. The normalized *n*-alkane peaks of BE-31 and BE-19 are compared in Figure 4A, with the differential in their relative concentrations shown below. In BE-19 the highest range homologs have been lost and there is an excess concentration that appears from about C_{21} to C_{26}, with some slightly enhanced lower members. This difference suggests that *n*-alkane biodegradation is beginning to diminish the highest members leading to a "bubble" effect, or downfield migration, of enhanced concentrations of newly formed lower MW *n*-alkane homologs.

This difference is further clarified by CSCIA of groups A–C (D has no residual *n*-alkanes). Averaged results for each group are shown in Figure 5. Here the compound-specific profiles for

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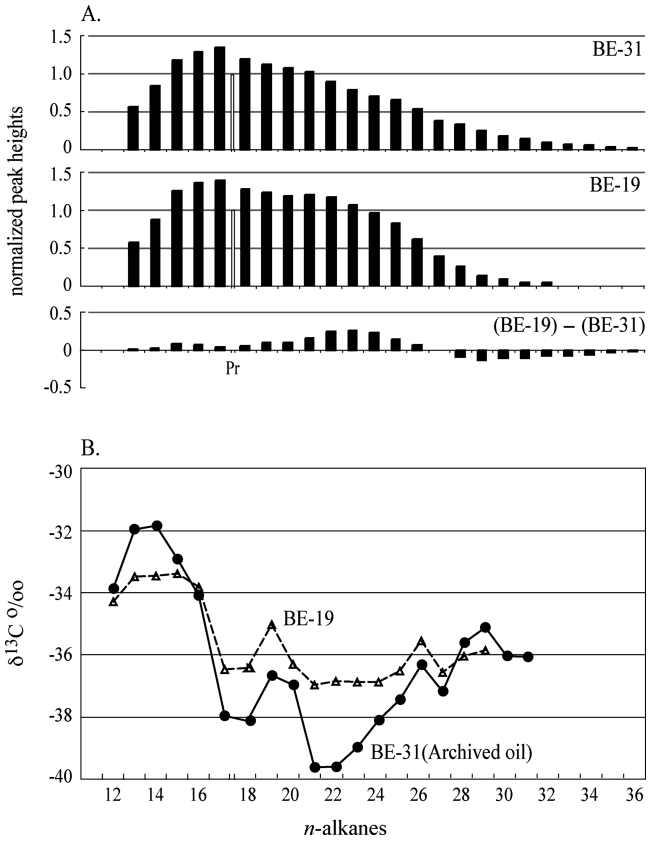


Figure 4. A) *n*-Alkane profiles of two Bemidji, MN, oil samples: BE-31 (unaltered oil) and BE-19 (lightly altered). Peaks were normalized to a conserved internal standard to allow comparison between samples. Differences between the two samples in the peak heights are at the bottom of this graph. B) Compound-specific carbon isotope profiles of *n*-alkanes in BE-31 and BE-19.

the three groups of progressively biodegraded samples are compared to unaltered oil, BE-31. This figure shows that *n*-alkanes in the three stages become progressively heavier, as with the Mandan oils, especially at the mid and lowest ranges. A significant difference appears approximately in the range of C_{21} to

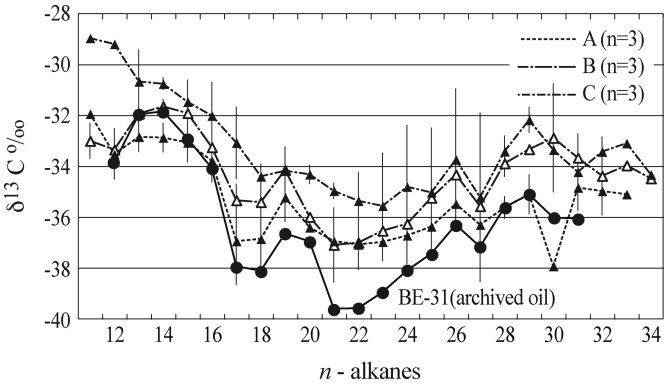


Figure 5. Compound-specific carbon isotope profiles of *n*-alkanes in the Bemidji, MN, oil samples. Samples were divided into four progressive stages of biodegradation, A–D (see text), then for simplicity, the profiles were averaged for each group. In the Bemidji grouping, the D-group lacks *n*-alkanes and therefore was excluded from the comparison.

C_{26}. In Figure 4B, the CSCI profile of BE-19 compared to BE-31 also shows that this range of compounds is the most enriched in ^{13}C, with some lesser enrichments of members with decreasing length of the homolog chain. This range of heavier carbon isotope values corresponds to the enhanced concentration range in BE-19 versus BE-31 in Figure 4A. Thus, in this sample there is evidence that the Bemidji biodegradation progression, just as in the Mandan progression, appears to proceed at least in part by higher end homologs being degraded to lower homologs, as evidenced by both enhanced concentrations and enrichment of new product in ^{13}C. Other Bemidji samples at stages of increased degradation show similar enrichments, although as degradation progresses, concentration enhancement is not as clear, suggesting that concurrent formation and degradation are occurring. Possibly some *n*-alkanes are being degraded as new *n*-alkanes of the same carbon number are being formed, and eventually the degradation outpaces the new *n*-alkane formation. In Bemidji samples there is not a buildup of enhanced concentrations of the lowest range, C_9 to C_{12} members, as was seen in the Mandan samples. However, that difference could be due to differences in laboratory work-up. Unlike diesel hydrocarbons that were analyzed by GC/MS as whole oils, the crude oil has been subjected to more evaporation steps because of being fractionated. However, C_{max} does move to lower MW alkanes with progressing degradation, just as with the Mandan samples. Alternatively, the degradation could be progressing fast enough in the latter stages of crude oil loss that all *n*-alkanes are being depleted faster.

A very high temperature GC analysis (data not shown) indicated only traces of *n*-alkanes or any other constituents greater than C_{36} in the unaltered Bemidji crude oil. This data indicates that excess *n*-alkanes are not derived from any very high MW constituents unobserved in regular GC/MS.

In the laboratory incubation experiment, normalized concentration profiles were calculated for *n*-alkanes undergoing methanogenic biodegradation in the North Sea crude oil (Bonnie Light, Chevron). Two replicates taken at each time period were averaged and are shown in Figure 6, comparing each time period with day 0. Changes in concentration relative to the starting concentration of each *n*-alkane are tabulated in Table 3. Biodegradation of *n*-alkanes progressed slowly with less than half of the *n*-alkane concentrations lost over the two-year period. Preferential loss from the mid- and high-range homologs is observable throughout the sampling period. However, enhanced concentrations in the low range are not obvious until the final sampling. In the 730-day samples both replicates include enhanced relative concentrations of 3% to 40% in the C_{10} to C_{14} homolog range. Although enhancement is not large, considering the time period of this experiment and the competing overall biodegradation and loss to the system, enhancement is significant. The laboratory methanogenic incubation, then, helps corroborate the progression of *n*-alkane biodegradation observed in the aforementioned field studies.

In summary, all three systems show a degradation progression with *n*-alkane losses from the higher end and show evidence of enhanced concentrations at lower ranges. In the field

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Table 3. *Remaining concentrations (relative to day 0) of* n*-alkanes in the crude oil during the laboratory methanogenic incubation.*

Carbon number	Active microcosms					Killed control				
	Day 0	Day 150[a]	Day 300[a]	Day 570[a]	Day 730[a]	Day 0	Day 150	Day 300	Day 570	Day 730
10	1.00	0.95, 0.93	0.94, 0.90	1.06, 0.98	1.33, 1.40	0.91	0.92	0.93	0.91	0.91
11	1.00	0.93, 0.95	0.94, 0.99	0.97, 1.09	1.21, 1.15	0.95	0.93	0.95	0.92	0.93
12	1.00	0.91, 0.90	0.92, 0.98	0.99, 1.01	1.16, 1.24	0.94	0.93	0.94	0.94	0.92
13	1.00	0.98, 0.92	0.95, 0.94	0.90, 1.02	1.03, 1.09	0.91	0.94	0.93	0.92	0.95
14	1.00	0.98, 1.05	0.90, 0.93	0.95, 1.02	1.08, 1.10	0.95	1.01	0.90	0.94	0.94
15	1.00	0.89, 0.94	0.87, 0.85	0.94, 0.99	0.88, 0.74	0.94	0.94	0.98	0.90	1.01
16	1.00	0.92, 1.03	0.86, 0.95	0.80, 0.72	0.64, 0.69	0.93	0.94	0.95	0.91	0.95
17	1.00	0.95, 0.99	0.81, 0.93	0.75, 0.80	0.55, 0.51	0.92	0.98	0.91	0.94	0.98
18	1.00	0.93, 0.97	0.79, 0.84	0.64, 0.73	0.45, 0.52	0.95	0.92	0.95	0.95	0.92
19	1.00	0.94, 0.98	0.84, 0.77	0.69, 0.58	0.60, 0.66	0.95	0.95	0.93	0.95	0.96
20	1.00	0.87, 0.95	0.77, 0.81	0.62, 0.65	0.57, 0.43	0.94	0.94	0.96	0.94	0.97
21	1.00	0.91, 0.99	0.82, 0.92	0.63, 0.67	0.56, 0.65	1.01	0.91	1.02	1.05	1.02
22	1.00	0.97, 0.93	0.89, 0.82	0.73, 0.68	0.76, 0.70	0.95	0.95	0.97	0.99	0.92
23	1.00	1.00, 0.90	0.90, 0.81	0.74, 0.66	0.79, 0.72	1.00	0.94	1.01	1.00	0.91
24	1.00	0.99, 1.03	0.80, 0.91	0.70, 0.75	0.75, 0.63	0.96	1.01	0.92	1.02	0.90
25	1.00	1.02, 0.96	0.80, 0.74	0.76, 0.69	0.69, 0.64	0.94	0.90	0.91	0.98	1.01
26	1.00	1.00, 0.95	0.81, 0.87	0.77, 0.68	0.66, 0.59	0.95	0.95	0.95	0.91	0.95
27	1.00	0.89, 0.85	0.78, 0.71	0.68, 0.55	0.59, 0.48	1.02	0.94	1.01	0.90	0.91
28	1.00	1.09, 1.05	1.07, 0.87	0.83, 0.64	0.64, 0.61	1.01	0.98	0.99	0.92	0.92
29	1.00	1.05, 1.06	1.00, 0.90	0.83, 0.75	0.55, 0.49	0.99	0.95	0.94	0.99	0.94
30	1.00	1.09, 1.07	0.94, 1.03	0.50, 0.64	0.35, 0.29	0.94	0.91	0.95	0.95	0.91
31	1.00	1.00, 0.93	0.78, 0.68	0.38, 0.47	0.28, 0.33	1.08	0.90	0.94	0.99	0.92
32	1.00	0.76, 0.74	0.57, 0.66	0.35, 0.32	0.25, 0.19	0.99	0.92	0.91	0.95	0.93
33	1.00	0.71, 0.65	0.48, 0.47	0.32, 0.29	0.22, 0.30	1.01	0.93	0.92	1.05	0.95

[a] Data includes replicate analyses from separate serum bottles at each time period (see Methods).

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studies, enhanced *n*-alkane concentrations are ephemeral since overall biodegradation would be competing with new alkane formation. However, several samples allow the progression to be teased out. In the Mandan samples (Figure 2) levels of *n*-C_9 through *n*-C_{12} are enhanced relative to the reference standard, starting with the freshest spill sample available, WF11, and including at progressively lower levels *n*-alkanes in downtown samples MW44, MW17 and PZ2. In the Bemidji site, instead of lowest-end buildup, we see a transitory appearance or "bubbling down through" of the enhanced *n*-alkanes at mid-range, which is particularly prominent in BE-19. Also, in the laboratory experiment a slowly developing low-range homolog enhancement can be observed.

n-Alkyl CHs, n-Alkyl Benzenes; Isorenieratane, A Branched-Chain Alkyl Benzene Series

The previous discussion of *n*-alkane progression is directly applicable to the *n*-alkyl CHs. *n*-Alkyl CHs are molecules with a cyclohexane ring attached to one end of a straight-chain alkane. Because the *n*-alkyl end of the molecule is more susceptible to biodegradation (see also Kim et al., 2005), the molecule appears to be preferentially attacked at its terminal end by the same process that shortens *n*-alkanes, resulting in progressively smaller *n*-alkyl CH homologs. Figure 7 shows extracted ion chromatograms of the *n*-alkyl CHs (m/z 83) of progressively degraded samples from both Mandan (A) and Bemidji (B). In the three Mandan samples, one can see that the main change is the developing "hump" of homologs near C_{10}. Superficially, this

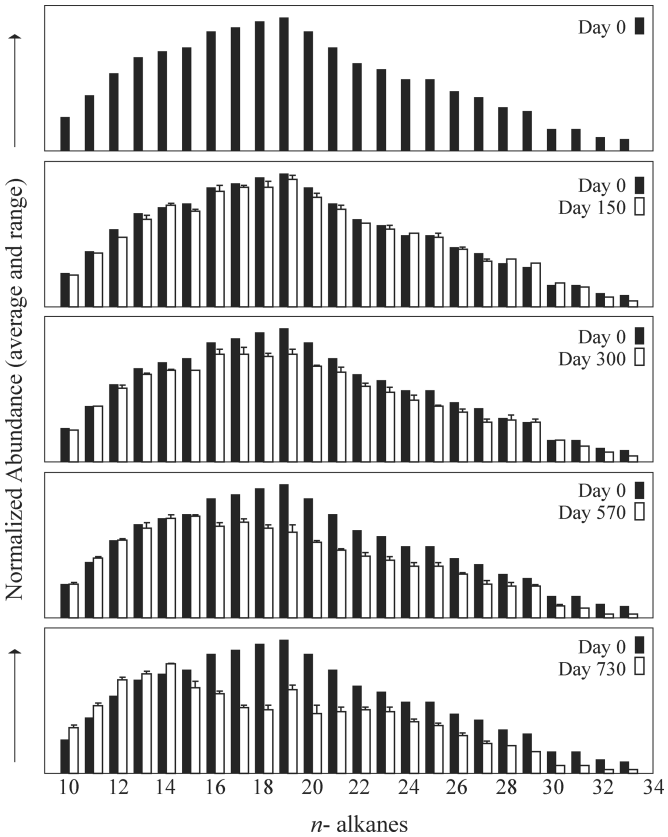


Figure 6. *n*-Alkane profiles for replicate runs of the laboratory incubation experiment. Tabulation is given in Table 3.

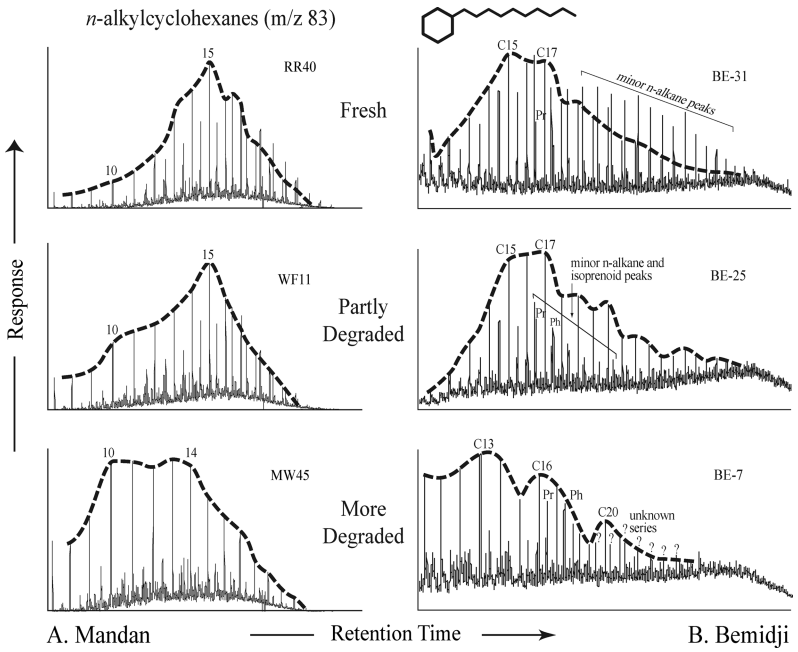


Figure 7. Extracted ion chromatograms (*m/z* 83, *n*-alkyl cyclohexane profiles) of three progressively biodegraded samples from A) Mandan, ND, and B) Bemidji, MN.

sample progression could look like a mixture of a lower temperature refinery product (the "hump") with remnants of diesel fuel. However, only downtown samples were argued to be co-contaminated (Stout et al., 2006), and the relatively recent WF11 with the C_{10} hump just beginning is strictly a railyard sample, separate from the downtown spill and therefore not impacted by any extraneous spillage. Three chromatograms of representative samples from both Mandan and Bemidji show losses from the high end with enhancement at the low end, similar to the *n*-alkanes.

Figure 8 gives a semi-quantitative look at this progression. This figure includes normalized members of homologous *n*-alkyl CHs with additional representative samples from the Mandan diesel spill (8A) and the Bemidji crude oil spill (8B). The same progression of high-end homolog loss occurs in both oils with the *n*-alkyl CH series (see also Hostettler and Kvenvolden, 2002). Because *n*-alkyl CHs are degraded more slowly than *n*-alkanes (Peters and Moldowan, 1993; Head et al., 2003), there is a lag time before substantial degradation occurs, as is seen in the Mandan samples in the similarity of the relatively fresh WF11 and EF1, the newer refueling areas with spills that are fresher than the downtown spill, to the unaltered RR40 (Figure 8A). However, even the WF11 and EF1 show the beginning of biodegradation in a progression very similar to that of the *n*-alkanes. There is some loss at the high MW end, along with formation of lower end homologs displaying an absolute concentration increase. This process then continues with progressively greater high-end loss and low-end enhancement, as reflected in the profiles of other older downtown samples, MW44, MW17, PZ2, and MW45.

Bemidji samples (8B) also show a lag time before substantial degradation occurs. Degradation stages A and B are not included in the figure because they show very little change relative to BE-31, the fresh crude oil. There is little degradation of *n*-alkyl CHs until *n*-alkanes are nearly depleted. At stage C (e.g., BE-28) there is some high-end loss and slight mid-range enhancement. At stage D (BE-6,7,8) there is significant high-end loss and significant low-end enhancement. Again, the logistics of the spill site affords no possibility of extraneous hydrocarbons entering the spill area to give these enhanced concentrations, nor is the system subject to differential hydrocarbon migration.

In addition, C_{max} in the distribution of both systems shifts to lower MW alkanes. The Mandan samples, with the least degraded sample (RR40) having a CH distribution of C_9 through C_{25} and a C_{15} maximum, were slowly changed to a distribution with a C_{10} maximum and enhanced C_{10} to C_{12} homologs. Bemidji samples exhibited a progression of lower end C_{13} and C_{14} members increasing while higher end C_{18} and C_{19} members decreased, with the maximum of the distribution shifting from C_{15} to C_{13} (Hostettler and Kvenvolden, 2002).

The degradation progression of a third homologous series, *n*-alkyl benzenes, at Mandan and Bemidji are shown in Figures 8C and 8D, respectively. Similar observations as the one mentioned above can be made for *n*-alkyl benzenes. Mandan samples (8C) display the same progressive high-end loss and low-end enhancement of homolog concentrations in spilled samples relative to RR40 undegraded diesel fuel. Bemidji samples (8D), with a somewhat irregular profile of *n*-alkylbenzenes, still display the same progressive loss from the high MW end of the distribution, after a similar lag time through the A and B degradation stages, as was seen in the aforementioned *n*-alkyl CHs. The Bemidji profile, however, does not manifest any low-end enhancement in the *n*-alkylbenzene suite, with all samples displaying similar

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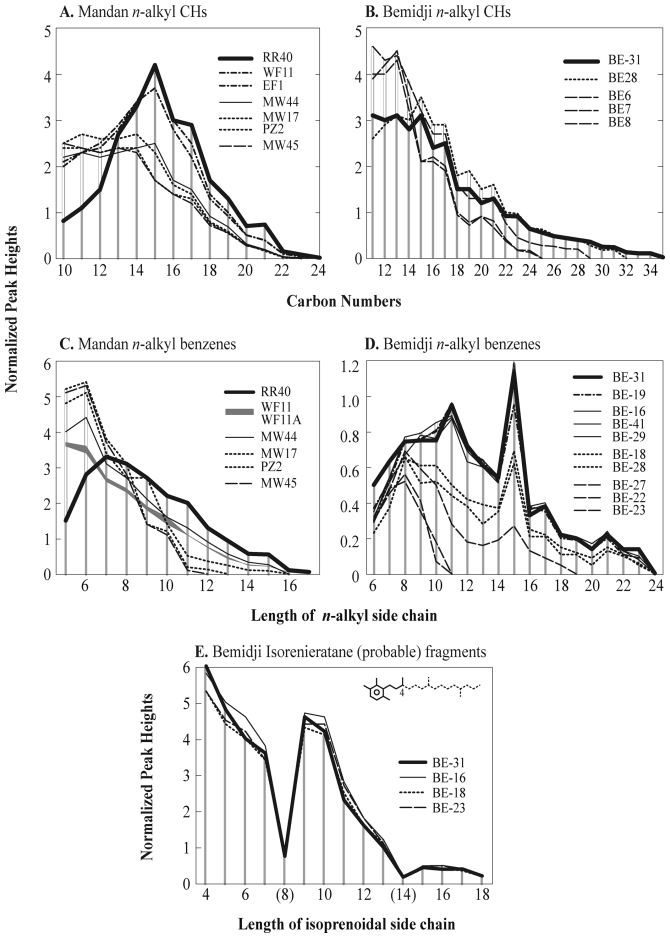


Figure 8. *n*-Alkyl cyclohexane and *n*-alkyl benzene degradation progressions in representative A and C) Mandan, ND, and B and D) Bemidji, MN, samples. E). For comparison the progression of substituted alkyl benzene homologs in the isorenieratane sequence is given. In each key the samples increase in biodegradative loss down the list.

levels at the low-end. This lack of low-end buildup could be due to relative rates of *n*-alkyl benzene degradation, where degradative loss is diminishing the suite faster than the low end can accumulate. However, the profile does display a clear downfield migration of C_{max} from C_{13} and C_{15} to about C_8. *n*-Alkyl benzenes, then, seem to have a very similar progression to the *n*-alkyl CHs.

One interesting comparison was made between the *n*-alkyl benzene series and another alkylated homologous benzene series prominent in Bemidji oil. Mass spectra of this series indicate that its components are characterized by three additional methyl groups on the benzene ring and an isoprenoidal side chain. The parent compound is most likely isorenieratane (C_{40}, mol. wt. 546), which gives rise to a series of C_{13}–C_{27} monoaryl fragments (having low chromatographic peaks at C_{17} and C_{23}; i.e., at points of branching on the isoprenoidal side chain). Structures (Figure 8E) are tentatively assigned from their mass spectra and by comparison with spectra given in a detailed study of these types of compounds by Koopmans et al. (1996). The degradation profile of these compounds is shown in Figure 8E and can be compared to that of *n*-alkyl benzenes in 8D. It has been reported in several studies (e.g., Volkman et al., 1984) that the presence of additional ring alkylation and branched alkyl chains significantly deters biodegradation. This is notably true in the Bemidji methanogenic system, as Figure 8E shows that the isorenieratane series is essentially untouched by the degradation that has almost completely removed the *n*-alkyl benzene series. The consistent lack of alteration of the isorenieratane family of compounds also gives further support to the premise that the spilled oil is not subject to differential migration, since the profile remains constant over a large MW range of its members.

These several series, then, illustrate that the *n*-alkanes are the earliest aliphatic series to be lost to biodegradation. Extrapolating from *n*-alkane patterns, it is apparent that long *n*-alkyl side chains on ring systems are uniquely susceptible to biodegradation and are being attacked at their non-ring terminus as if they were alkanes (see also Kim et al., 2005). Branching on the long alkyl side chain and/or on the ring significantly retards biodegradation. Enhanced levels of lower MW members as degradation progresses are apparently intrinsic to the process, especially with *n*-alkanes and *n*-alkyl CHs.

Forensic Fingerprint of the Mandan Oil

The preceding sections showed that aliphatic oil constituents with long-chain *n*-alkyl side chains are in a state of flux due to biodegradation throughout the time period of sample spillage. Therefore, more refractory constituents must be used for forensic fingerprinting. The fingerprint can then be applied both to evaluate whether there is a common source for the spilled LNAPL and whether the downtown spill might be contaminated with lower temperature refinery products other than railroad diesel. With respect to source, the report by Hostettler et al. (2001) concluded, as previously discussed, that the narrow range of bulk carbon isotope values, similar PAH patterns, and a common red dye marker, all indicated a common source for all the spilled oils (i.e., the off-road diesel fuel used by the railroad and supplied by the local refinery). Additional evidence is provided here. The report included quantitative analysis of the three major PAH families common to most oils, naphthalene, phenanthrene, and dibenzothiophene, and their C_1 to C_3 homologs. Figure 9 summarizes data from all the samples, including downtown samples (straight lines), railyard samples and the reference off-road diesel (dashed lines), and one unrelated diesel standard (Restek, long-dashed line). The figure shows that there is remarkable consistency in profiles of downtown and railyard samples. There is one exception among the downtown samples. One naphthalene line is different from the others. This analysis represents sample FR4, a very shallow sample possibly with some air exposure on the periphery of the spill, which was noted in the report to have parameters indicating physical weathering (i.e., loss of lower MW constituents). This sample falls back into line in the plots of higher MW PAHs, the phenanthrenes and dibenzothiophenes. The unrelated Restek diesel standard is seen to be

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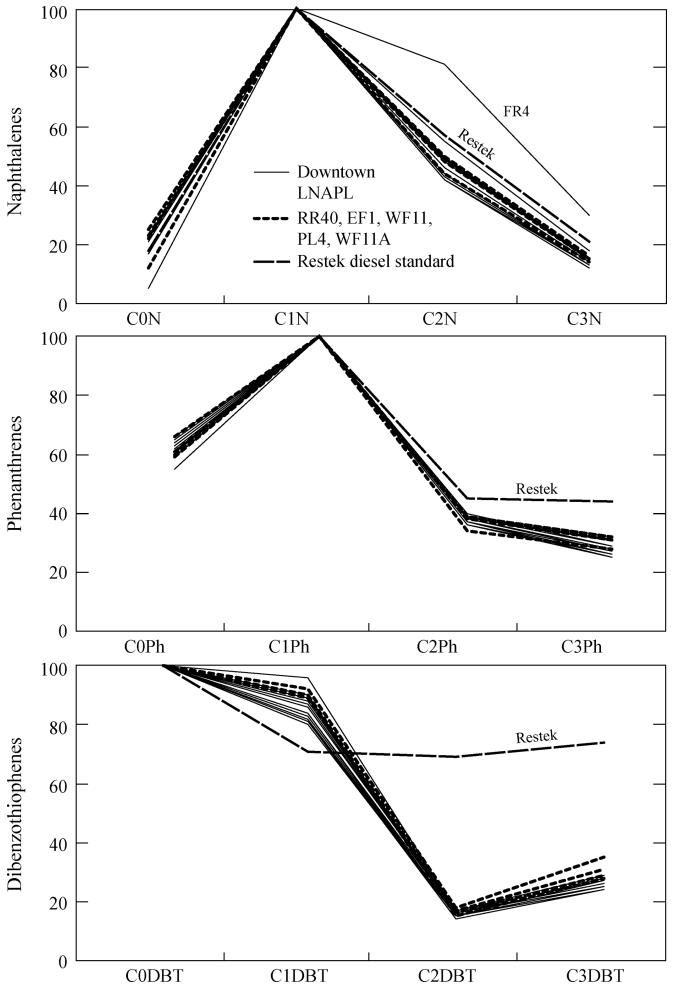


Figure 9. Summary of the PAH quantitation profiles on all Mandan samples, compared to an unrelated Restek diesel standard (data from Hostettler et al., 2001).

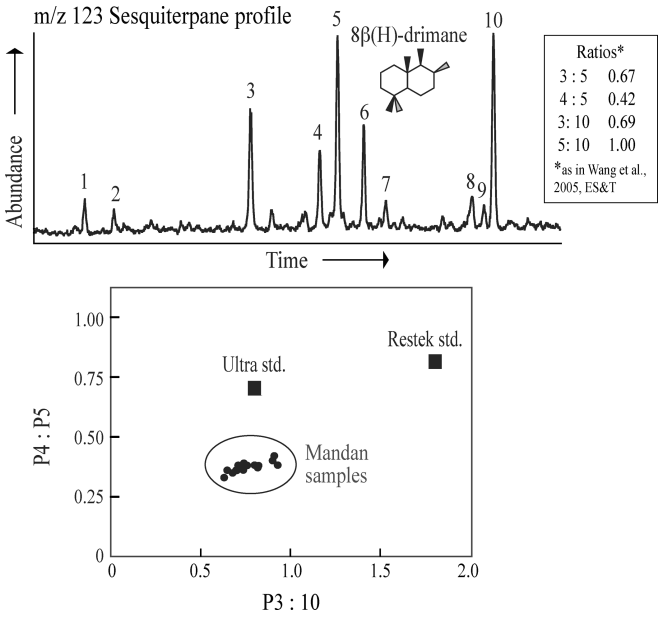


Figure 10. A typical extracted ion chromatogram at m/z 123 of the drimane sesquiterpane series, with a cross-plot of suggested ratios to compare the Mandan, North Dakota, samples with unrelated diesel standards.

dissimilar to Mandan samples, particularly in the dibenzothiophene profiles.

Another source parameter suggested in recent literature (Wang et al., 2005) was applied to the samples, namely a cross-plot of various drimane-related biomarkers. Figure 10 contains the drimane distribution in the Mandan oils and shows how they cluster together, especially in contrast to two unrelated purchased standards. We also did a cross-plot of C2 and C3 dibenzothiophene/phenanthrene parameters (widely used as part of oil fingerprints, Bence et al., 1996; see Figure 11). Although there is some small scatter in these ratios for various Mandan samples (Hostettler et al., 2001), the figure shows that sites still cluster quite closely, especially considering the range of values shown by other unrelated samples. This graph supports the observation made in the report that small differences in the chemical constituents reflected in these ratios do not represent different sources, but can be explained simply by small differences possible in the crude oil supplied by the refinery over the 50 years of railroad use.

One other parameter that proved useful in correlating Mandan samples is the pristane/phytane (Pr/Ph) ratio, another commonly used source parameter. The narrow range for the refinery oils and oil products (1.2 to 1.4) brackets all the spill samples (Table 4). This data also has relevance in the discussion of admixture below.

All these different contributors to the forensic fingerprint support the argument for a common source for all the spilled oils. However, even with evidence for a common source for the downtown and on-site railroad diesel samples, admixture with other refinery fuels is a different problem. The range and distribution of constituents in, for example, kerosene, on-road diesel, heating oil, and gasoline (the only likely co-contaminants in this environment) are significantly different, and many correlation parameters are not applicable between different products. Of this list of

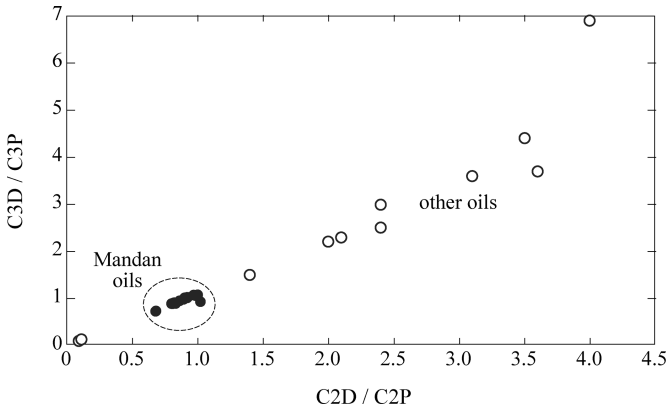


Figure 11. A cross-plot of the C2 and C3 dibenzothiophene/phenanthrene source parameters showing the Mandan, North Dakota, oils and other unrelated diesel fuel and oil samples.

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Table 4. *Ranges of pristane/phytane (Pr/Ph) and C_4 benzene/drimane (C_4Bz/Dr) ratios in Mandan samples (RR40, WF11) mixed with kerosene (KER)*

Field Samples		Pr/Ph	C_4Bz/Dr
Downtown samples		1.2–1.3	4.6–6.0
East & West Fueling (EF, WF), PL4		1.2–1.3	6.0–6.6
RR40		1.4	
Crude oil used in the refinery		1.2	
Mixed samples	Percent	Pr/Ph	C_4Bz/Dr
RR40 and Kerosene mixtures:			
RR40	100	1.4	
RR40:KER	90:10	1.6	
RR40:KER	75:25	1.7	
RR40:KER	50:50	1.9	
KER	100	3.1	
WF11 and Kerosene mixtures:			
WF11	100		6.0
WF11:KER	90:10		6.9
WF11:KER	75:25		8.9
WF11:KER	50:50		10.8
KER	100		17.3

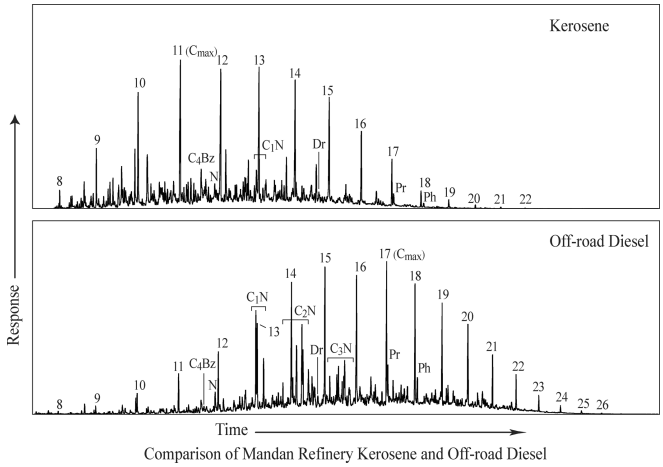


Figure 13. Gas chromatography/mass spectrometry chromatograms of Mandan, ND, refinery; kerosene and off-road diesel (RR40) numbers indicate *n*-alkane homologs; N and C_{1-3}N indicate the PAH naphthalene and C1 to C3 homologs, respectively; Pr = pristane; Ph = phytane; C_4Bz = a tetramethylbenzene; Dr = drimane.

fuels it has already been noted that one shallow outlying site had traces of a chemical marker of gasoline, namely iso-octane. But this site represented a very minor amount of the total spillage and no other site contained evidence of gasoline. Heating oil was hardly used in the downtown area since heating was done early on using coal, then the area transitioned to natural gas. On-road diesel might be possible, but the railroad did not focus on this fuel. In any case, our quantitation of PAHs in RR40 off-road diesel versus an on-road diesel from the same refinery gave a profile that showed differences, especially in the phenanthrene series (Figure 12). As for kerosene, however, the enhanced *n*-alkyl CH levels displayed earlier did give rise to the possibility of admixture with that fuel. However, alkyl CH profiles alone are not adequate to prove or suggest admixture. In a mixture there would be significant redistribution of oil components and not just an enhancement of lower MW *n*-alkyl CHs.

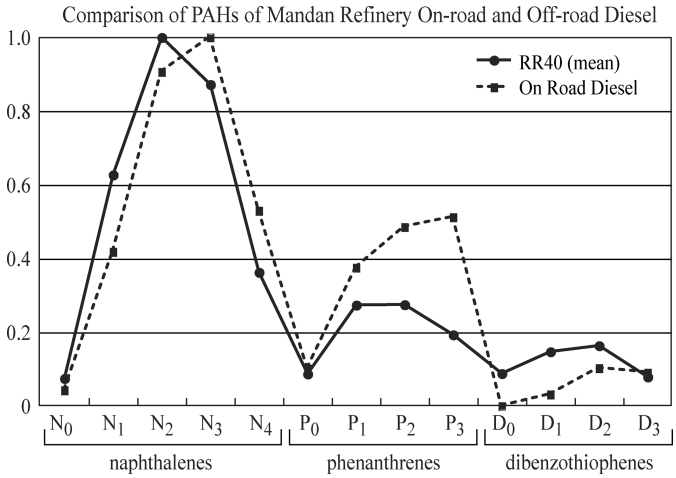


Figure 12. Quantitation values for the PAH profiles of RR40. (off-road diesel) and a sample of Mandan, ND, refinery on-road diesel.

Chromatograms of the RR40 and a sample of kerosene also from the local refinery are displayed in Figure 13. The range of constituents in the two fuels differs considerably. Therefore, many of the ratios in the fingerprint used for source determination are not relevant to this problem. However, one ratio, the simple pristane/phytane ratio (Pr/Ph), is particularly useful. Figure 13 shows that for off-road diesel the pristane and phytane peaks are near mid-range of the constituents, and thus the relationship of these two isoprenoids is not changed in the refinery process from the values in the crude oil (1.2 to 1.4). The refinery kerosene, however, contains phytane near the cutoff at the high temperature end of its distillation range. Therefore, Pr/Ph in the kerosene has a diminished denominator, thus giving it a significantly higher value (3.1). Mixtures of kerosene with RR40 (Table 4) indicate that from 50 to 10% kerosene in off-road diesel, the Pr/Ph value of the solutions is still higher than that of the initial oil. Therefore, if kerosene is a co-contaminant in the LNAPL, it would have to be at quite a low level.

A second comparison was made to help evaluate if admixture with kerosene shifts undegraded constituent levels, this time using a spilled sample mixed with refinery kerosene. WF11 was chosen because it is strictly an on-site railyard sample, presumably not subject to the co-contamination claimed for the downtown samples. Two highly refractory constituents common to both the spilled fuel and the refinery kerosene were chosen, a C_4-benzene and drimane, and their ratios (C_4Bz/Dr) in the pure oil, pure kerosene, and in mixtures calculated. Table 4 gives the values for WF11 with 50%, 25%, 10% and 0% kerosene. Chromatograms of these mixtures (data not shown) clearly show that *n*-alkyl CHs increase significantly at the low end of the distributions, as expected, with the added kerosene and that the Cmax shifts lower also. The ratio of C_4Bz/Dr is 6.0 in the pure oil and 17.3 in the kerosene, with intermediate values for the mixtures. However, when one compares the range given in Table 4 for all

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the downtown samples, these values are all either at the same level as pure WF11 or somewhat lower. There is no evidence of admixture with kerosene when non-degrading constituents are used to evaluate the system.

In summary, co-contamination of off-road diesel by other fuel entities can be evaluated using specific markers (gasoline), variable PAH profiles (on-road diesel), and carefully selected parameter ratios (kerosene). This analysis supports the value of the usefulness of a preponderance of evidence versus using only one line of evidence. Except for a trace of gasoline in one sample, other fuel contaminants in the downtown Mandan oil spill samples were not indicated, in sharp contrast to claims made by railroad scientists (Stout et al., 2006; Stout and Uhler, 2006).

Conclusions

This study looks carefully at forensic fingerprint parameters applicable to two long-term methanogenic oil spills. *n*-Alkyl CHs, whose range in different refinery products is frequently used in forensic work to define these products, are shown in the Mandan off-road diesel spill not to be useful because they are changing due to biodegradation over the time period of spillage. Anomalous enhanced levels of lower MW *n*-alkyl compounds are suggested to be attributable to methanogenic biodegradation progression and not to extraneous hydrocarbon admixture. Rather than CHs, profiles of more refractory PAH and biomarker compounds such as the drimane series and the pristane/phytane ratio work quite well to establish commonality of the spilled LNAPL samples and to investigate possible mixed contamination. In evaluating three biodegrading long-chain *n*-alkyl series, we obtained more evidence for the proposed progression of biodegradative loss of *n*-alkanes and *n*-alkylated CH and benzene homologs in the methanogenic oil spill systems of Mandan, North Dakota, and Bemidji, Minnesotta. A laboratory simulation of biodegradation under methanogenic conditions supports the progression of loss of *n*-alkanes seen in these field studies. *n*-Alkyl compounds are apparently attacked at the end of *n*-alkane chains, resulting in the formation of progressively lower MW homologs that are isotopically enriched in ^{13}C. This unique progression is different from what is commonly found in the literature, where degradation is frequently seen to occur initially and greatest from the low MW end of the distribution. Enhancement in concentrations of lower range homologs is reported for the first time in these systems and should be considered in similar studies. This degradation process impacts not only *n*-alkanes, but also specific families of compounds with long *n*-alkyl side chains. The mechanism of this methanogenic progression within the non-polar environment of the oil body is unknown and requires additional research.

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